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Research on Distributional Compositional Categorical Model in Both Classical and Quantum Natural Language Processing

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***Abstract*—Language models based on neural network primarily focus on learning patterns and representations from the input data, without explicitly modeling the grammatical structure of the text, thus often lacks of the consideration of syntax information. The distributional compositional categorical (DisCoCat) model is a framework that can both capture the distributional information of words and the syntactic structure of sentence. Thus, we research the application of DisCoCat models in both quantum and classical natural language processing. Specifically, in classical and quantum natural language processing, we respectively converted sentences into tensor networks and quantum circuits based on the DisCoCat model to get representation containing both semantic and syntactic information of language in classical and quantum computer. We conduct experiment for a text binary classification task in both classical and quantum natural language processing setting to verify the effectiveness of DisCoCat model. The experimental results shown that the combination of DisCoCat with quantum computing has significant potential.**

Keywords—DisCoCat, QNLP, text representation

I. Introduction

Neural network including recurrent neural networks (RNNs) [1], convolutional neural networks (CNNs)[2], and transformer[3] have shown remarkable success in natural language representation learning. However, one of the limitations of these models is their inability to effectively capture syntactic information, such as grammar rules and sentence structures, which are crucial for language understanding. The Distributional Compositional Categorical (DisCoCat) model [4] has emerged as a powerful framework in language representation for integrating semantic and syntactic information, allowing for the generation of network features that capture both semantic and syntactic aspects of text. In classical natural language processing, DisCoCat model can be used to generate text representation in tensor network [5] form, which reflect the distributional and compositional nature of language. Besides, in quantum language processing [6]. Quantum language model aims to leverage the analogy between computational linguistics and quantum mechanics to enhance the representation and processing of language data. One promising approach in QNLP is the integration of quantum computing with DisCoCat models, which providing an approach to represent text as quantum circuits that can be processed by quantum computer.

In order to probe the actual effectiveness of DisCoCat model in text representation across classical and quantum NLP, we conducted binary text classification experiments in both classical and quantum NLP settings. In the classical NLP setting, we developed a method based on the DisCoCat model to represent text as tensor networks and validated it through experiments on classical computers. In the quantum NLP setting, we proposed a method to represent text as quantum circuits using the DisCoCat model and conducted quantum text classification experiments through classical computer simulations. The experimental results shown that DisCoCat is advantageous in quantum NLP, opening new research opportunities for efficient and accurate text processing

II. Related Work

As the basic language unit, word representation and sentence representation are crucial part of natural language representation. Distributional word embedding techniques map words to dense vectors in a continuous vector space, which posits that words appearing in similar contexts tend to have similar meanings. Universal word embeddings include Word2Vec [7], GloVe [8], ELMo [9], and BERT [10], but a universally accepted technique is lacking. Current sentence- embedding methods can be divided into three group: (1) Word collection models [11], (2) Pragmatic coherence methods [12] and (3) Semantic comparison models [13]. These categories are selected based on how they reflect the authors' under- standing of the relationship between words and sentences, as well as the roles sentences play in human language. Recently, QNLP is mainly consider the various NLP challenges such as lexical semantic ambiguities, semantic composition, information retrieval, and text classification. Quantum algorithms [14] are designed for quantum computers, while quantum-inspired or quantumlike models [15] are classical algorithms that draw inspiration from quantum principles. The term "quantum NLP" encompasses both classical NLP

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algorithms influenced by quantum physics and quantum algorithms used for NLP tasks

III. Preliminaries

A. DisCoCat Model

In modern natural language processing, the vector space model is commonly utilized to determine the meaning of individual words. Words with normalized vectors are deemed similar if the inner product of their vectors is close to one, a method known as distributional semantics. However, this technique is limited in its ability to analyze longer text units like phrases and sentences. To address this limitation, compositional semantics algorithms are employed to derive the meaning of larger text units by considering the meanings of individual words and their grammatical structure. The distributional compositional semantic model (DisCo) integrates both distributional and compositional approaches to incorporate grammatical structure into the composition of word vectors.

In this model, each grammatical type is assigned a tensor product space based on Lambek's pregroup grammar. For example, the representation of a noun is a vector $\vec{n}$ in space N. Transitive verbs' representations $\vec{v}$ are in space N ⊗ S ⊗ N, where S is the meaning space of sentence. Representation vector of an adjective $\vec{adj}$ is in space $N \otimes N$. The diagrammatic notation of vectors, tensors and linear maps is shown in Fig.1. The two way of composing word tensor is sequential and parallel, whose diagrammatic notations are shown in Fig.2.

B. Quantum Computing

A qubit can be represented as a linear combination of the basis states $|0\rangle$ and $|1\rangle$, denoted as:

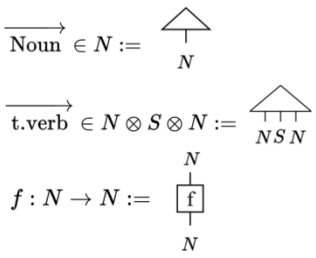


Fig. 1. Diagrammatic notation of vectors, tensors and linear maps

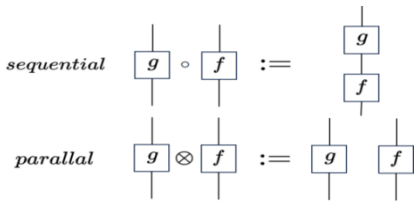


Fig. 2. Diagrammatic notation of sequential and parallel composing

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1)$$

where α and β are complex numbers satisfying the normalization condition $|\alpha|^2 + |\beta|^2 = 1$. This represents the qubit being in a superposition of the $|0\rangle$ and $|1\rangle$ states. This superposition property allows qubits to represent both 0 and 1 simultaneously, enabling quantum computers to perform complex calculations in parallel. In addition, Qubits entanglement property enables quantum computers to perform certain calculations much faster than classical computers. Entanglement between two qubits can be described using the Bell state, such as:

$$|\phi^+\rangle = \frac{(|00\rangle)+|11\rangle}{\sqrt{2}} \quad (2)$$

Quantum gates are used to manipulate qubits, changing their states to perform quantum computations. The design and optimization of quantum gates play a crucial role in advancing quantum computing technology and achieving efficient quantum computations. Quantum circuits represent a sequence of quantum gates applied to qubits to perform quantum computations. A suitable quantum circuit can optimize the efficiency of the algorithm.

IV. Methodology

We propose a four-way methodology to construct the transformation process of mapping the text into tensor networks or quantum circuits (shown in Fig.3): (1) syntactic analysis, (2) string diagram generation, (3) diagram rewriting, (4) tensor network / quantum circuit transformation. For comparison, we also generate the tensor network and quantum circuit representation of text based on bag-of-word and word-sequence models through the same procedure. We use sentence" The lazy cat always curls up in the soft mat" as example for following demonstration.



Fig. 3. Transformation procedure

A. Syntactic Analysis

We use syntactic analysis to extract the grammatical structure of a sentence to create a syntax tree. This involves breaking down the sentence into its constituent parts (such as nouns, verbs, adjectives, etc.) and determining how these parts relate to each other based on the rules of the language's grammar. The syntax tree visually represents the hierarchical structure of the sentence, with each node representing a word or phrase and the connections between nodes indicating the relationships between them.

B. String Diagram Generation

Given the syntax tree of a sentence generated by syntactic analysis process, we need to convert it into a string diagram. In string diagram, words are represented by boxes, and syn- tactic information flow through the wires. An example string diagram

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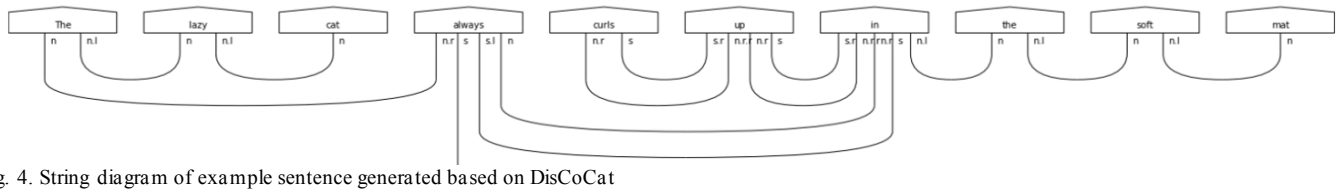


Fig. 4. String diagram of example sentence generated based on DisCoCat

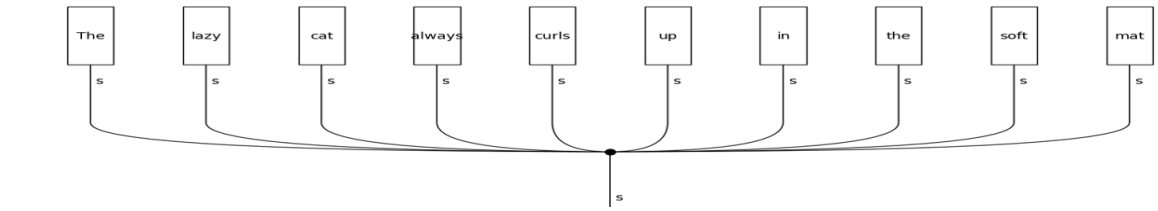


Fig. 5. String diagram of example sentence generated based on bag-of-word

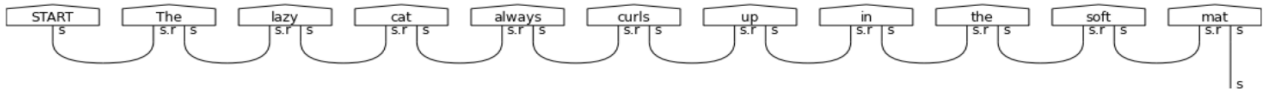


Fig. 6. String diagram of example sentence generated based on word-sequence mode

of sentence generated based on DisCoCat model is shown in Fig.4, where "s" represents a node of sentence type, n" represents a node of noun type, "l" represents a node of preposition type, "n.r" represents a node of noun modifier, "s.r" represents a node of sentence modifier, "n.r.r" represents a modifier of a noun modifier, such as an adjective or adverb modifying a noun modifier, and "n.l" represents a preposition modifying a noun. Bag-of-word model treats the words in a sentence as an unordered set, without considering their syntactic relationships. It then maps each word to a specific word vector and combines these vectors according to certain rules to generate a vector representing the sentence. The word-sequence model constructs string diagram of words based on their order of appearance in the sentence, thus taking into account the sequential information of words. This model can capture richer semantic relationships between words, as adjacent words may influence each other's meanings. Examples of bag-of-word and word-sequence model are shown in Fig.5 and Fig.6.

C. Diagram Rewriting

In order to simplified syntactic derivations in pregroup form to avoid complexity that could lead to excessive use of hardware resources and long training times, we introduce diagram rewriting process. By applying simple rules such as merging duplicate elements or eliminating unnecessary elements in diagram rewriting process, the number of nodes in the string diagram can be significantly reduced. After the diagram rewriting process, the string diagram in Fig.4 can be reconstructed as Fig.7.

For comparison the rewritten bag-of-words string diagram focuses solely on word frequency, while the sequence model considers only the order of words. Therefore, rewriting a generated string by connecting words in a graph would be simpler, but it would not be able to recognize the semantic relationships between words, and no words would be removed during the rewriting process.

D. Tensor Network / Quantum Circuit Transformation

In this step, we convert abstract string diagram representation to a concrete tensor network or quantum circuit by applying parameterization techniques known as ansatz. An ansatz serves as a mapping that determines various choices, such as the number of qubits associated with each wire in the string diagram and the specific parameterized quantum states corresponding to individual words.

In classical natural language processing scenario, string diagram need to be converted into tensor network for use in subsequent NLP tasks. The key point in converting a string diagram into a tensor network is to assign dimensions to each atomic type in the diagram. This involves parameterizing a Tensor Ansatz instance with dimensions for the noun space and the sentence space. The wires of the diagram are annotated with these dimensions, indicating a concrete tensor network as the result. Tensor network converted from string diagram based on DisCoCat model is shown in Fig.8.

In quantum natural language processing scenario, string diagram need to be converted into quantum circuit for processing on quantum computer. In order to generate the quantum circuit, we define the number of qubits for all atomic types that occur in the diagram – in this case, for the noun type and the sentence type. The quantum circuits obtained from processing rewritten string diagram using different representation models are shown in Fig.9(a), (b), and (c) respectively.

The main differences among the three circuit diagrams lie in their depth and specific gate sequences. It can be found that the DisCoCat model, with its multi-level structure, can better capture the semantics and contextual information of sentences. Circuits generated based on the word-sequence model requires more instructions to represent the arrangement of words. In contrast, circuits generated based on the bag-of-words model only consider word frequencies, neglecting word order and semantic relationships. As a result, circuits generated using the

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bag-of-words model is relatively simple, but that limits the understanding of semantics and contextual information.

V. Experiment

We conducted a text binary classification task (food or IT) to test the effectiveness of text representation based on the DisCoCat model in both quantum and classical NLP settings. We split the dataset into three parts: training set, validation set, and test set. Here are five samples from the training set: ['skillful person bakes dinner.', 'skillful woman bakes meal.', 'woman runs useful application.', 'man bakes tasty meal.', 'person prepares tasty meal.'].

A. Experiment Setting

In both quantum and classical NLP setting, completing a text binary classification task involves two stages: preparation and training.

a) Preparation: The preparation stage involves the following main processes: (1) reading text data from the dataset and performing preprocessing, including tokenization and generating syntax trees; (2) converting the text into a string diagram; (3) rewriting the generated diagram to simplify the model complexity; (4) converting the optimized diagram into tensor network or quantum circuit.

b) Training: Once tensor net work or quantum circuits are created, we can train language model to solve the binary text classification task. Here's the main process of training quantum classifier: (1) Instantiate the model using the created quantum circuits; (2) Define loss and evaluation metric, in this setting we use standard binary cross-entropy as the loss; (3) Initialize the trainer by providing model, loss, optimizer, evaluate function et al.; (4) use trainer to train the quantum classification model.

The training process of classical and quantum classifier based on DisCoCat, bag-of-word and word-sequence representation models are shown in Fig.10 and 1 respectively. Ideally, in quantum experiment setting, the preparation stage is completed on a classical computer, while the training stage is completed on a quantum computer. However, due to constraints, the training stage of this experiment is simulated in classical environment.

B. Result and Analysis

In order to better understanding the performance of the model at different stages and the binary classification task accuracy on the test set, the trends of loss values and accuracy for both the training and validation sets are given in this section.

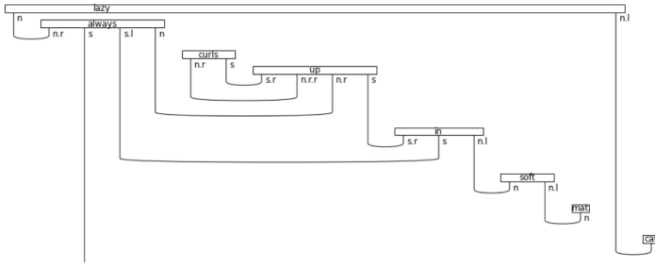


Fig. 7. Rewrited string diagram based on DisCoCat representation

The comparison graph of quantum NLP experiment setting is shown in Fig.10. The bag-of-words model achieves the best t-

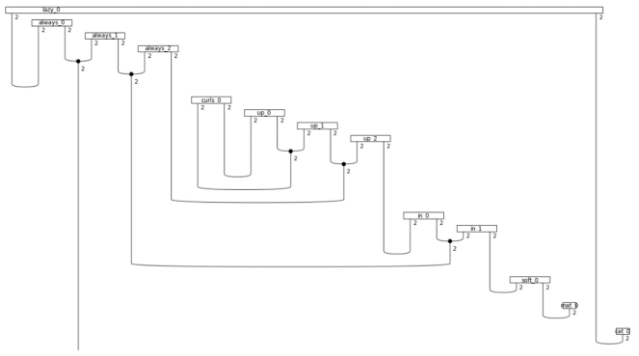
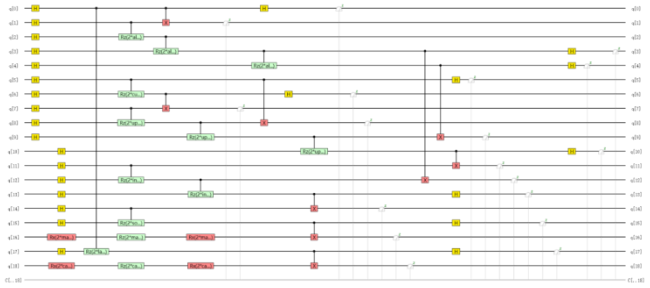
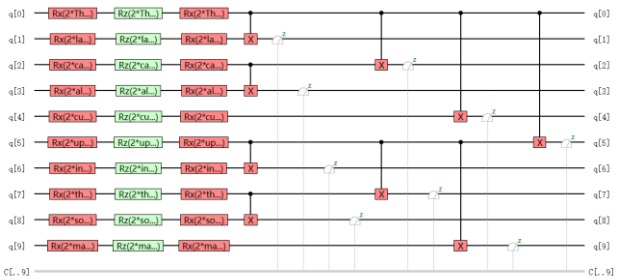


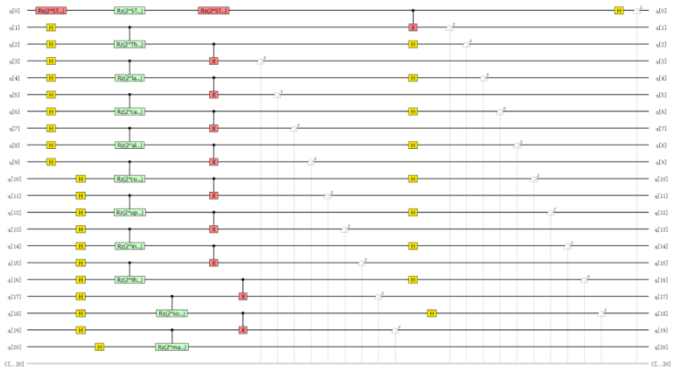
Fig. 8. DisCoCat tensor network converted from string diagram



(a) DisCoCat



(b) bag-of-word



(c) word-sequence

Fig. 9. Quantum circuit converted from string diagram

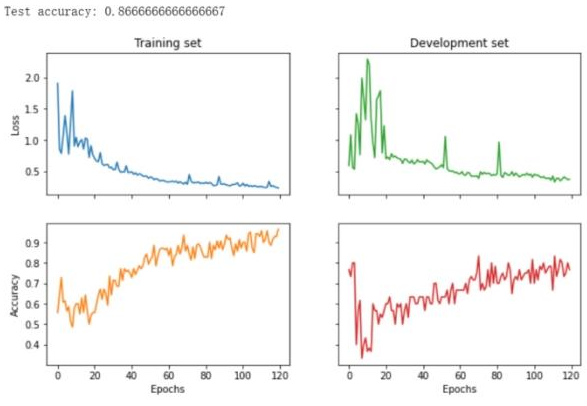
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raining and testing accuracy, reaching 1.0. Although the DisCoCat model is not as effective as the bag-of-words model, it shows faster decreases in loss values and improvements in accuracy during training, with less fluctuation in both loss values and accuracy. Considering that the effectiveness of the bag-of-words model may significantly decrease when using richer and more complex corpora, and that the bag-of-words model may not perform well in handling word order, grammar structure, and semantics, the DisCoCat model is more suitable for applications in quantum natural language processing. As shown in Fig.11, in classical NLP setting, the DisCoCat model achieves the best training and testing accuracy, reaching 1.0. The loss values for both the training and validation sets gradually decrease with the number of iterations, while the bag-of-words model and the word-sequence model increase.

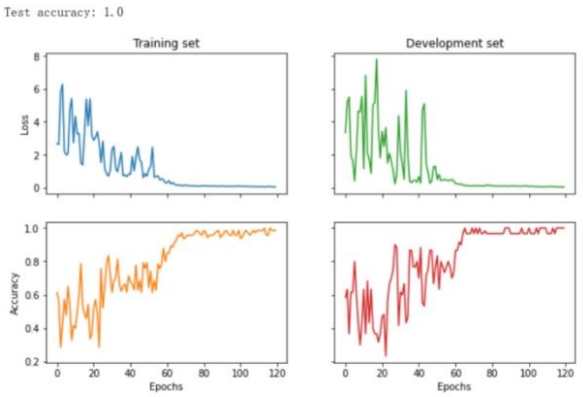
In addition, in terms of final accuracy, the DisCoCat model also performs the best in training. Overall, as the number of iterations increases, it is found that using quantum natural language processing methods for training results in relatively low loss values for the training and validation sets, while accuracy continues to improve. In terms of reducing loss values and improving accuracy, the word bag model and word sequence model show the most significant improvement in training performance. Therefore, they have certain advantages in reducing the loss values of the training and validation sets and improving accuracy.

VI. Conclusion

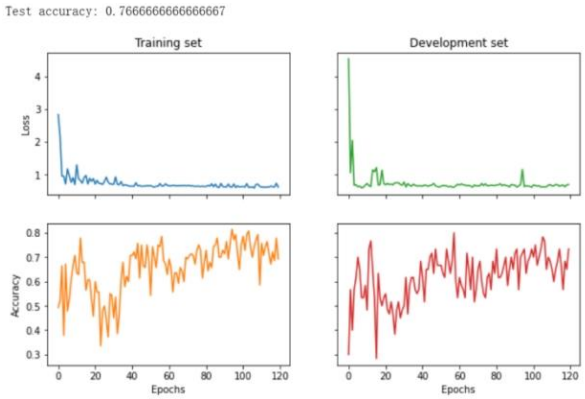
In this paper, we study the effectiveness of DisCoCat model for text representation which encode the syntax information of language in both quantum and classical natural language processing. In quantum NLP scenario, we develop a procedure to mapping text to quantum circuit which can be processed by quantum computer. In classical NLP scenario, we develop a procedure to mapping text into tensor network. The result of experiment on binary text classification task shows the potential of DisCoCat model in text representation in both quantum and classical NLP setting compared to other representation approach such as bag-of-word and word-sequence model.



(a) DisCoCat

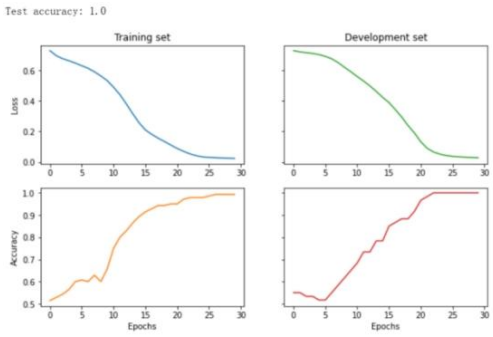


(b) bag-of-word

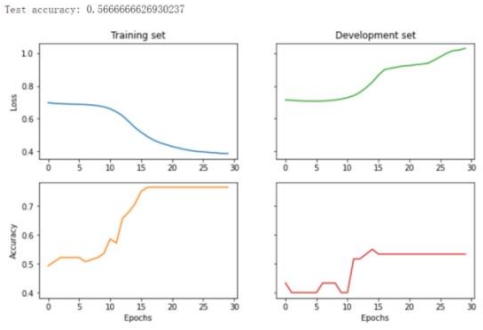


(c) word-sequence

Fig. 10. Training effect in quantum setting

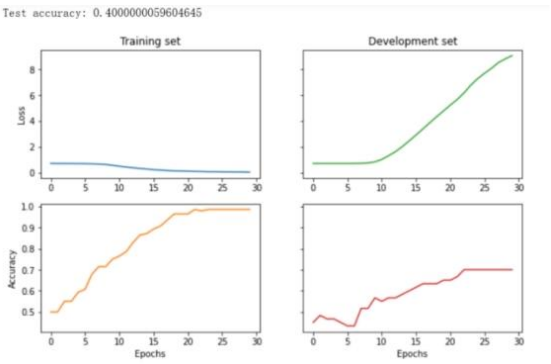


(a) DisCoCat



(b) bag-of-word

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(c) word-sequence

Fig. 11. Training effect in classical setting

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